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17 November 1961
SUPERSEDING
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MILITARY SPECIFICATION

WELDING MACHINES, ARC, DIRECT-CURRENT, VARIABLE VOLTAGE TRANSFORMER-RECTIFIER TYPE

1. SCOPE

1.1 Scope. - This specification covers welding machines of the direct current, transformer-rectifier design for shipboard use.

1.2 Classification. - The welding machines shall be of the following types, as specified (see 6.1):

Type I - Stationary.
Type II - Portable.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-692 - Steel Sheets, Carbon, Cold Rolled.
QQ-S-693 - Steel. Sheet, Hot-Rolled.

MILITARY

MIL-C-25 - Capacitors, Fixed, Paper-Dielectric, Direct-Current (Hermetically Sealed in Metallic Cases).
MIL-S-901 - Shockproof Equipment, Class HI (High Impact), Shipboard Application, Tests for.
MIL-C-915 - Cable, Cord and Wire, Electrical (Shipboard Use).

FSC 3431

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MILITARY (cont'd)

MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-E-986 - Electrodes, Welding, Mineral Covered, Low Hydrogen-Ballistic Plating and Armor Weld Application.
MIL-E-2036 - Enclosures for Electric and Electronic Equipment (Naval Shipboard Use).
MIL-R-15109 - Resistors and Rheostats, Shockproof (Naval Shipboard Use).
MIL-R-15137 - Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-S-15291 - Switches, Rotary, Snap Action, Bureau-Design.
MIL-E-15599 - Electrodes, Welding (Medium Steel).
MIL-S-16113 - Steel Plate, Hull and Ordnance, Structural, Black (Uncoated) and Zinc-Coated (Galvanized).
MIL-P-16298 - Preservation, Packaging, Packing and Marking of Electric Machines Having Rotating Parts (Includes Associated Repair Parts).
MIL-O-16485 - Ohmmeter (Insulation, Resistance-Indicating).
MIL-W-16878 - Wire, Electrical (Insulated, High Temperature).
MIL-M-17059 - Motors, Alternating-Current, Fractional HP (Shipboard Use).
MIL-C-17361 - Circuit Breakers, Air, Electric, Insulated Enclosure (Shipboard Use).
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-R-18281 - Rectifiers, Metallic: Selenium, Copper Oxide, and Magnesium Copper Sulfide Types, Naval Shipboard Use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-248 - Qualification Tests for Welders.

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DRAWINGS

BUREAU OF SHIPS
9-S-5259-L - Rapid Coupling for Electric Cables.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Qualification. - The variable voltage direct current arc welding machines furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein and has been listed on or approved for listing on the applicable qualified products list.

3.2 Definitions. -

3.2.1 Setting. - A setting is defined as a point of adjustment of the welding machine controls.

3.2.2 Volt-ampere curve. - The volt-ampere characteristic of any setting is a plot of the terminal voltage versus output current for that setting when an attached load is varied to obtain an output current from zero to short-circuit current.

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3.2.3 Normal current. - Normal current for any setting is the steady state current which will flow at that setting when an external connected load is adjusted to give terminal voltages within the applicable arc voltage ranges specified in table I.

Table I - Range of arc voltage.

Amperes	Arc voltage
25 to 50	18 to 25
50 to 100	18 to 27
100 to 150	20 to 29
150 to 200	21 to 30

3.2.4 Normal output. - Normal output at any setting is the power output in watts when the machine is operating against the normal current for that setting.

3.2.5 Rated voltage. - Rated voltage for the machine is the voltage specified in table II.

Table II - Size of welding machines.

Size	Rating of machine		Minimum range of output current (amperes) from 20 volts to rated voltage
	Rated current in amperes	Rated voltage in volts	
200	200	30	40 to 250

Note. - All ratings shall be based on continuous welding at rated load on a 60 percent duty cycle.

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3.2.6 Rated current. - Rated current for the machine is the load specified in table II.

3.2.7 Rated output. - Rated output for any size machine is the normal output when the machine is operating at rated voltage and rated current.

3.2.8 Maximum open-circuit voltage. - The maximum open-circuit voltage is the open-circuit voltage at the output terminals of the machine when the machine is set for maximum output with the machine operating at no load and when the machine is connected to a power source which will supply rated voltage to the welding machine.

3.2.9 Current. - Unless otherwise specified herein, current is the output current of the machine.

3.2.10 Voltage. - Unless otherwise specified herein, voltage is the output terminal voltage of the machine.

3.2.11 Arc voltage. - Arc voltage refers to the voltage drop which exists across the arc during the welding operation. Arc voltage is dependent upon the size and type of electrode used, the current and the length of the arc. Arc voltage shall be considered to be generally within the range of values specified in table I.

3.2.12 Duty cycle. - A welding machine with a 60 percent duty cycle is defined as one which will operate continuously under duty cycles consisting of six minutes at rated output or four minutes at no load.

3.2.13 Dripproof protected enclosure. - A dripproof protected enclosure is defined as one meeting the dripproof protected enclosure requirements of Specification MIL-E-2036 except as follows:

(a) The enclosure shall be so designed and constructed that drops of liquid or solid particles, falling on the enclosure from an angle not greater than 15 degrees from the vertical, cannot interfere with the satisfactory operation or eventually result in unsatisfactory operation of the equipment within the enclosure.
(b) The enclosure tests required by Specification MIL-E-2036 do not apply.

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3.3 General design characteristics. -

3.3.1 Apparatus comprising a machine. - The welding machine shall be completely assembled, compactly mounted and shall comprise all equipment necessary for furnishing electric energy to the welding circuit. The welding machine enclosure shall be designed to meet the dripproof protected enclosure requirements as defined in 3.2.13.

3.3.2 Safety. - Particular care shall be taken in the design and construction of welding sets to insure safety to the operating personnel from danger of electrical shock or moving parts.

3.3.3 Material. - The material, including insulation for terminal boards, shall conform to Specification MIL-E-917.

3.3.4 Welding machine shall conform to the class HI shock requirements of Specification MIL-S-901.

3.3.5 Welding machines shall be designed and constructed to withstand the environmental vibration (type I) conditions specified in Standard MIL-STD-167.

3.3.6 Weight and dimensions. - Welding machines shall be as compact and lightweight as practical consistant with the strength and reliability as required herein except that the machines shall not exceed 340 pounds for type I and 365 pounds for type II. The overall dimensions of the welding machines shall be such as to allow the machine to pass through a 25 inch diameter hatch.

3.3.7 Ambient temperature. - Welding machines shall be designed for an ambient temperature of 40°C.

3.3.8 Blower motors. - Blower motors shall conform to service C of Specification MIL-M-17059 except that motors shall meet the shock requirements of 3.3.4 when mounted in the welding machine. Motors shall be of the following classification:

Service - C.
Ambient temperature - Special (see 3.3.13).
Voltage - 220 volts or 110 volts.
Duty - Continuous.
Enclosure - Totally enclosed.

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Horsepower - As required to provide adequate cooling for the welding machine.
Speed - Constant.
Phase characteristics - Single phase, 60 cycle.
Type of bearing - Ball or sleeve.
Class of insulation - Class A, B or H.

3.3.9 Rectifiers. - Rectifiers shall conform to Specification MIL-R-18281 except as follows, and shall be of the following classification:

Type of rectifier - Type S or C.
Rating - As required for the size of welding machine involved.
Ambient temperature - Special (see 3.3.13).
Duty - Intermittent; 6 minutes with welding machine delivering rated load, 4 minutes with welding machine at no load.
Circuit - As required for the welding machine.
Type of cooling - Blower cooled.
Maximum permissible plate operating temperatures for rectifiers.

Type of rectifier	Maximum plate operating temperature °C.
Copper-oxide (type C)	60
Selenium (type S)	75

3.3.10 Resistors and rheostats. - Resistors and rheostats shall conform to Specification MIL-R-15109 and shall conform to the following classification:

Form - EW or IW.
Ambient temperature - Special (see 3.3.13).
Insulation - Class B or C.
Duty - Intermittent or continuous as required by the particular application of rheostats and resistors in the welding machine when the machine is continuously delivering rated load at a 60 percent duty cycle.
Mounting - As required by the design of the welding machine.
Application - As required by the design of the welding machine.

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3.3.11 Transformers and reactors. - Transformers and reactors shall be manufactured in accordance with Specification MIL-E-917 and shall be designed for the ambient temperature specified in 3.3.13.

3.3.12 Capacitors. - Capacitors shall be in accordance with Specification MIL-C-25 for direct current (d.c.) circuits and as approved by the bureau or agency concerned for alternating current (a.c.) circuits.

3.3.13 The ambient temperature for motors, rectifiers, resistors, rheostats, transformers, reactors and capacitors shall be considered as the maximum ultimate temperature of the cooling air supplied to each of these parts when the welding machine is continuously operating under an ambient temperature of 40°C. and when the machine is continuously delivering rated load at a 60 percent duty cycle.

3.3.14 Suitability of operation. - The welding machine shall be suitable for welding with electrodes conforming to Specifications MIL-E-15599 and MIL-E-986. The characteristics of the set shall be such that an operator qualified in accordance with Standard MIL-STD-248, may easily strike and maintain a stable arc and satisfactory weld at any suitable setting for the electrode used within the specified range of the machine without losing the arc more than once per minute..

3.3.15 Power service. - Welding machines shall be designed to operate from a 440 volt or a 220 volt (60 cycle, a.c., r.m.s.) three phase power source and shall be readily reconnectable for either voltage. Further, the welding machines shall be designed to perform in accordance with the requirements herein at any input voltage of not more than 10 percent above or below rated voltage and at any frequency of not more than 5 percent above or below rated frequency. Welding machines shall be connected for use on a 440 volt power supply prior to shipment.

3.3.16 Operating characteristics. -

3.3.16.1 Current adjustment. - Current within the range specified in table II shall be obtained from the terminals of the welding machine at any voltage between 20 volts and rated voltage inclusive. There shall be no setting (a setting suitable for the electrode being used) at which it will be difficult to weld. Current shall be adjustable between the minimum and maximum values specified in table II in not less than 30 increments. Not more than two controls shall be used for current adjustment. At rated voltage, the increase in current from any setting to the next higher setting shall not exceed 5 percent of the current at the lower setting.

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3.3.16.2 At any setting corresponding approximately to rated output, starting with the welding machine and components operating at room temperature (25° + 10°C.), the current shall not vary more than plus or minus 15 percent for the machines, from the initial current during a 6 minute run against a resistive load adjusted to maintain rated voltage. The input voltage to the machine shall be held at a constant value and the adjustments of the machine shall not be changed during this run.

3.3.16.3 Open circuit voltage. - The maximum open-circuit voltage shall not exceed 100 volts under any operating conditions.

3.3.16.4 Volt-ampere characteristics. - Volt-ampere characteristic curves shall show that at all points below rated voltage, the voltage decreases with an increase in current. No point of the curve shall indicate a current higher than that for short-circuit. The machine shall have such characteristics that for any setting, at any arc voltage from 20 volts to rated voltage inclusive, a 10 percent change in arc voltage shall not result in more than a 15 percent change in current.

3.3.16.5 Ripple voltage. - At any machine setting and at any load between no load and maximum specified current, the peak to peak value of the output ripple voltage shall not exceed 20 percent of the average value of the output voltage.

3.3.16.6 Life. - Welding machines shall be designed for a minimum life expectancy of 5000 hours under the conditions of moisture encountered aboard ship, the conditions of vibration and heat specified herein and with the machine operating continuously under duty cycles consisting of:

(a) Six minutes with the machine delivering normal current at any setting between the minimum specified output and the specified rated output settings.
(b) Four minutes with the machine operating at no load.

3.3.17 Short-circuit and recovery characteristics. - Welding machines shall conform to the short-circuit and recovery characteristics specified in 3.3.17.1 through 3.3.17.2 which shall be determined after the welding machine and its components are operating at temperatures corresponding to the values attained at the completion of the temperature rise tests specified in 4.5.5.

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3.3.17.1 Short-circuit current. -

3.3.17.1.1 When starting with open circuit and then shorting the welding terminals of the welding machine, the peak or momentary short-circuit current for any setting shall not exceed 550 percent of the normal current for that setting.

3.3.17.1.2 When starting with normal current for any setting and then shorting the welding terminals, the peak or momentary short-circuit current for any setting shall not exceed 400 percent of the normal current for that setting within the specified range of operation.

3.3.17.1.3 The steady state short-circuit current for any setting shall not be greater than 250 percent nor less than 110 percent of the normal current for that setting.

3.3.17.1.4 The transient current shall not drop below 60 percent of the average normal current for any setting when a short-circuit is opened through a resistance giving normal current for that setting.

3.3.17.1.5 The permissible time of recovery from short-circuit current to within 5 percent of normal current for any setting shall not exceed 0.5 seconds when a short-circuit is opened through a resistance giving normal current for that setting.

3.3.17.2 Recovery voltage. - On any setting, the minimum recovery voltage, when a short-circuit is opened through a resistance giving normal current for that setting as required by 3.3.17.1.5, shall not be less than 70 percent of the final load voltage for that setting.

3.3.18 The maximum no load currents shall not exceed 2.0 amperes at 440 volts, or 4.0 amperes at 220 volts.

3.3.19 Maximum permissible temperature rises. - The temperature rises of welding machines shall not exceed the values shown in individual equipment specifications referenced herein and shown in table III when determined on basis of rated output of the welding machine and on a 60 percent duty cycle. Temperature rise measurements shall be taken by method 1, 2 or 3 of Specification MIL-E-917.

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Table III - Maximum permissible temperature rises.

Item	Part of machine to be measured	Method of measurement	Class A insulation		Class B insulation		Class H insulation	
			For 40°C. ambient	For 50°C. ambient	For 40°C. ambient	For 50°C. ambient	For 40°C. ambient	For 50°C. ambient
			°C.	°C.	°C.	°C.	°C.	°C.
1	Single layer insulated windings or uninsulated windings in contact with insulation	Method 1	55	45	75	65	130	120
		Method 2 or 3	65	55	85	75	145	135
2	Multilayer windings	Method 1, 2 or 3	50	40	70	60	120	110
		Method 2	60	50	80	70	125	115
3	Bare copper not in contact with insulation	Method 1	80	70	80	70	150	140
4	Bearings (see Note 1)	Method 1	50	40	50	40	110	100
5	Core or other mechanical parts in contact with insulated parts	Method 1 or 3	55	45	75	65	130	120
6	Contacts:	Method 1						
	Laminated copper, solid contacts		50	40	50	40	50	40
			60	50	60	50	60	50
	Solid contacts, silver faced		75	65	75	65	75	65

Note 1. - Maximum bearing temperatures shall not exceed 90°C. except as follows: Where the use of class H insulation requires the bearing to operate at temperatures in excess of 90°C. but not more than 150°C., silicone grease shall be used and the bearing shall be heat stabilized.

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3.3.19.1 The temperature rise of items 1 and 2 of table III shall be measured by method 1 (thermometer and thermocouple), method 2 (resistance) or method 3 (embedded detector). The determination of the values to govern as obtained by these methods shall be taken as specified in 3.3.19.1.1.

3.3.19.1.1 If the winding resistance is less than 0.01 ohm at 25°C. thermometer or thermocouple temperatures (method 1) shall apply and if the winding resistance is more than 0.01 ohm at 25°C. the highest values of temperatures obtained by thermometer or thermocouple (method 1) and resistance (method 2) shall apply. Embedded detector temperature values (method 3) are acceptable and may be used in lieu of either of above applications.

3.3.20 Efficiency. - The efficiency of welding machines shall not be less than the applicable values shown in table IV and shall be determined in the manner specified in 4.5.8.

Table IV - Efficiency values.

Percent of rated output	200 ampere machine percent efficiency
25	60
50	62
75	62
100	58
125	50

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3.3.21 Dielectric strength. -

3.3.21.1 Before assembly in the welding machine, the insulation of various components shall withstand dielectric strength tests required by the specifications for such components referenced herein except that rectifiers and capacitors (other than capacitors used for motor starting) shall withstand a dielectric test voltage of twice rated voltage plus 1000 volts but not less than 1500 volts for a period of one minute. Transformers and reactors shall withstand a test voltage of twice the rated voltage of the winding plus 1000 volts but not less than 1500 volts for a period of one minute between the winding and ground and between the winding and all other windings on the same core. Before assembly in the welding machine, transformer and reactors shall be subjected to an induced potential of twice normal voltage which shall be impressed across each winding of transformers and reactors for a duration of one minute. All dielectric test voltages shall be 60 cycle, a.c., r.m.s. except that the induced potential test voltage shall be not less than 120 cycles, a.c., r.m.s.

3.3.21.2 The completely assembled welding machine shall be designed to withstand a dielectric test voltage of twice rated voltage of the circuit plus 1000 volts but not less than 1500 volts (60 cycle, a.c., r.m.s.) between circuits and between each circuit and ground. When the welding machine incorporates the use of a blower motor, the motor and motor capacitor (if used) shall be disconnected when dielectric tests are being conducted on the welding machine. A separate dielectric test shall be conducted on the motor and capacitor. Such dielectric tests shall meet the requirements for the motor specification referenced herein.

3.3.22 Audio noise. - The welding machine shall operate without objectionable audio noise at all loads within the applicable load ranges specified in table II. Any degree of noise which is comparably greater than that which is inherent in any given type or design of machine shall be considered objectionable.

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3.3.23 Parallel operation. - Welding machines shall operate in parallel with other machines of the same type and design so that the difference in current supplied by any one machine from that supplied by any other machine will not exceed 10 percent of the rated current of one machine. The load division of 3.3.22 applies:

(a) At any voltage between 20 volts and rated voltage and at all total loads within the applicable range of output current shown in table II multiplied by the number of machines in parallel. For example, if two 200 ampere machines are to be paralleled, the range of output current for one machine would be 40 to 250 amperes as shown in table II. Therefore the range of total load supplied by the two 200 ampere machines in parallel would be from 80 amperes to 500 amperes.
(b) After the indicators on the controls of all machines to be paralleled have been set to the same value.

3.3.24 Lifting means. - Welding machines shall be provided with lugs, eyes, holes, handles or hand rails suitable for lifting the machine.

3.3.25 Welding control. - The welding control shall include contact making devices, rheostats, switches and other equipment necessary to provide adjustments for the range of current and voltage specified in table II. All welding controls shall be mounted in a dripproof protected enclosure as defined in 3.2.13. All welding controls shall be easily accessible at convenient height, and shall be of such design that they can be readily operated by an operator wearing a welder's glove. Welding terminals shall be protected from accidental contact or short circuit. All other live parts shall be contained within the welding control enclosure. All adjusting devices shall be provided with calibrated plates designed to show the magnitude of current for each setting.

3.3.26 Reactance. - The welding machines shall include reactance in the welding circuit which shall be effective for all welding conditions.

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3.3.27 Output terminal connections. - Welding terminals shall be of the stud and wing nut type and shall be fitted with female angle couplings conforming to type FA-1 on Drawing 9-S-5259-L or equivalent type couplings satisfactory to the bureau or agency concerned. Terminals and fittings shall be copper, brass or bronze of ample conducting capacity for rated current. Welding terminals shall be provided with a suitable identification plate to show the connections for operation of the machine with "straight" (electrode negative) or "reverse" (electrode positive) polarity.

3.3.28 Inclined operation. - Welding machines shall be designed to operate satisfactorily when inclined at an angle of 30 degrees in any direction from the normal horizontal position.

3.3.29 Input connections. - The input power shall enter the welding machine through a 4-pole polarized receptacle and plug conforming to the construction specified herein.

3.3.29.1 The connection terminals of the receptacle shall be identified by visible lettering or coloring (black, white, red and green) on a contrasting background.

3.3.29.1.1 The connection terminals marked "Black", "White" and "Red" shall be used to supply power to the welding machine. The connection terminal marked "Green" shall be used to ground the enclosure of the welding machine.

3.3.29.2 The plug and receptacle shall be designed in such a manner, that, after complete wetting with sea water and subsequent activation by a 440 volt power supply (60 cycle, a.c., r.m.s.), no leakage current nor voltage to ground should be observed on any portion to be grasped by the hand of an operator for the purpose of connecting or disconnecting the welding machine.

3.3.30 Switch or circuit breaker. - The primary or input circuit to the welding machine shall contain a manually or magnetically operated circuit breaker or manually operated switch which shall be so connected as to open all input power circuits to the welding machine. The switch shall conform to Specification MIL-S-15291. The circuit breaker shall meet the requirements of Specification MIL-C-17361. The rating of the switch or circuit breaker shall be such as to permit operation of the welding machine on 220 and 440 volt power sources.

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3.3.31 Internal wiring. - All connecting wiring shall be clamped into position to prevent abrasion of the insulation. All wire shall conform to Specification MIL-W-16878. The type and size of wire shall be such that it is of sufficient capacity, temperature and voltage rating for the intended duty.

3.3.32 External wiring. - External permanent wiring on the welding machine shall conform to the insulation requirements of Specification MIL-C-915 and shall be enclosed within flexible metal conduit clamped in position to prevent abrasion of insulation.

3.3.33 Terminal boards. - Terminal boards shall be constructed of insulation material conforming to the requirements specified herein. Such insulation material shall be supported by steel panels or framework. All terminal boards shall be easily accessible.

3.3.34 Diagram of connections. - A diagram of connections shall be attached to the inside of the cover of the welding machine or inside the welding control panel cover of the machine. Such diagram of connections shall be protected by a coating of transparent varnish or other suitable material.

3.3.35 Painting. - Painting shall be accomplished in accordance with the requirements for application of primer and enamel specified in Specification MIL-E-917.

3.4 Special devices. - Special devices designed to add to the economy, facility and convenience of operation shall be furnished when specified (see 6.1). The suitability, effectiveness, durability and trouble free operation of the special devices shall be demonstrated by means of tests. The design of special devices shall be subject to approval by the bureau or agency concerned. In the event the devices are approved, a description of the devices and the results of tests shall be included in the test report.

3.5 Accessories. - The following accessories shall be furnished with each machine when specified (see 6.1):

(a) One fully insulated electrode holder of good commercial design and suitable capacity.
(b) Fifty feet of grounding cable type TRF, size 105 and fifty feet of electrode cable type TRXF, size 105 conforming to Specification MIL-C-915.
(c) Power supply cable shall be furnished as specified (see 6.1).

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3.6 Type I machines. - Type I machines shall be provided with a base suitably drilled for installation of holding-down bolts.

3.7 Type II machines. - Type II machines shall be provided with suitable truck or running gear supports for bolting to the truck frame. However, if the truck frame is of rugged construction, the supports may be omitted. Portable units of the machine shall be equipped with skids and carrying handles in lieu of wheels. Skids shall be so constructed as to support a load equal to four times the normal loading of the parts without breakage or permanent set in any part. Machines with skid mounting shall be designed to be inherently capable of remaining upright (mechanically stable) when tilted 30 degrees in any direction from the normal horizontal position.

3.8 Repair parts and tools (onboard). - Onboard repair parts and tools shall be furnished, when specified (see 6.1), in accordance with Specification MIL-R-15137 and shall be based on the total number of such welding machine parts of each size, type and manufacture installed in the welding machines for each ship and shall be supplied as specified in table V.

Table V - Repair parts required.

Item	Name of part	Quantity of each part to be furnished per ship based on the number of identical installed parts or sets of identical parts per ship		
		1 to 5	6 to 10	Over 10
1	Blower motor (complete with fan)	1	1	1
2	Blower motor brushes (sets)	2	4	6
3	Capacitor (each size and type)	1	2	2
4	Rectifier stack (each size and type)	1	2	2
5	Resistor (each size and type)	1	2	2
6	Movable contacts for each size and type of rheostat (sets)	1	2	2
7	Springs for each size and type of rheostat (sets)	1	2	2

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Table V - Repair parts required (cont'd).

Item	Name of part	Quantity of each part to be furnished per ship based on the number of identical installed parts or sets of identical parts per ship		
		1 to 5	6 to 10	Over 10
8	Complete switches (each size and type)	1	2	2
9	Input receptacle and plug	1	1	1
10	Output terminal connectors (set)	1	2	3

Note. - The word "set", as used in table V, denotes a quantity equal to the number of each item installed in a complete welding machine.

3.9 Identification and information plates. -

3.9.1 Identification plates shall be attached to the part of the machinery or equipment which will not ordinarily be renewed during its normal service life, and shall be located in a readily accessible position where they can be read at all times without danger to personnel. Material shall be able to withstand the same environmental conditions as the item to which the plate will be attached.

3.9.2 Where an identification or information plate is applicable to a specific part or adjustment on the welding machine, the plate shall be installed in close proximity to the part or adjustment to which it refers. In general, the plate shall be mounted directly over or directly below the part or adjustment.

3.9.3 Plates shall be mounted by means of cadmium plated machine screws, nuts and lock washers or by means of self tapping screws. Where self tapping screws are used, they shall be of sufficient length to completely penetrate the enclosure to which the plate is to be attached and the point of the screw shall not endanger personnel or electrical equipment.

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3.9.4 Markings on identification plate for complete welding machine. - The data marked on the identification plate for the complete welding machine shall include the following:

(a) Manufacturer's name, identification symbol and serial number.
(b) Government contract number.
(c) Maximum open circuit voltage.
(d) Power service (volts, phases, frequency and so forth).
(e) Duty cycle rating.
(f) Output rating (volts and amps).
(g) Current range.
(h) Blank space for Government inspector's official stamp.
(i) Bureau or agency stock number.

3.10 Technical manuals. - The technical manuals shall be manufacturer's commercial manual furnished with covers substantial enough to withstand normal handling. These shall be modified, if necessary, to include assembly, interconnection, set-up or installation, operation and repair parts of the unit. They shall contain all information necessary for trouble shooting and normal maintenance. Two copies of the manual shall be supplied with each welding machine unless otherwise specified (see 6.1).

3.11 When any specification which forms a part of this specification requires that the product be subjected to and pass qualification tests, only products which are listed on the applicable Qualified Products List on the date of invitation for bids or which may be added to that Qualified Products List subsequent to that date shall be utilized in the construction of equipment specified to be in accordance with this specification. In the event no Qualified Products List has been issued, the contractor shall request instructions as to what testing will be required to determine whether the product meets the requirements of the specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Qualification tests[1]. - Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. These tests shall consist of the tests specified in 4.5.

4.3 Sampling for lot acceptance tests. -

4.3.1 Inspection lot. - All welding equipment of the same type and size presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.3.2 Sampling for selective tests. - Sample machines shall be selected from each lot in accordance with inspection level L-8 of Standard MIL-STD-105, and each of these units shall be subjected to the tests specified in 4.4.2.

4.3.3 Sampling for periodic tests. - Sample machines shall be selected from each lot in accordance with table VI and each of these machines shall be subjected to the tests specified in 4.4.3.

Table VI - Sampling for periodic tests.

Number of machines in lot	Number of units to be tested
1 to 200	1
201 to 500	2
501 and over	3

4.4 Lot acceptance inspection. -

4.4.1 Routine inspection. - Each unit shall be subjected to the following routine inspection:

Inspection	Paragraph
General inspection	4.5.1
Noise	4.5.6
Special devices	4.5.11

[1]Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.2.1).

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Inspection	Paragraph
Dielectric tests	4.5.12
Insulation resistance	4.5.13
No load and rated load input power	4.5.14

4.4.2 Selective tests. - Sample machines selected in accordance with 4.3.2 shall be subjected to the following tests:

Test	Paragraph
Range of output current	4.5.2.2
Current adjustment measurements	4.5.3.3

4.4.3 Periodic tests. - Sample machines selected in accordance with table VI shall be subjected to the following tests:

Test	Paragraph
Temperature tests	4.5.5
Short circuit and recovery tests	4.5.7
Efficiency measurement	4.5.8
Inclined operation	4.5.10
Parallel operation	4.5.15
Current adjustment measurements	4.5.3.1 and 4.5.3.2
Weight	4.5.9
Range of output current	4.5.2.1

4.4.4 Rejection of units. - Failure of welding machines to conform with specification requirements for any test shall be counted as a defect and the welding machine shall be rejected. If the number of nonconforming machines in any lot exceeds the quantities shown in table VII, the lot shall be rejected.

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Table VII - Number of nonconforming machines determined from results of routine, selective and periodic tests.

Number of welding machines in inspection lot	Routine tests	Selective tests	Periodic tests
40 and under	0	0	0
41 to 200	1	0	0
201 to 500	2	1	0
501 and over	3	2	0

4.5 Test procedure. -

4.5.1 General inspection. - Each unit shall be subjected to a thorough inspection to ascertain that the workmanship and design conform to this specification. The fit of parts shall be observed with particular emphasis on inspection with regard to interchangeability of parts which are likely to require replacement during the normal service life of the unit.

4.5.2 Range of output current. -

4.5.2.1 Tests shall be conducted so that volt-ampere curves may be plotted for four settings corresponding to maximum specified current at rated voltage, rated current at rated voltage, minimum specified current at 25 volts and minimum specified current at 20 volts. Readings shall be taken at points corresponding to no load, at short circuit and at least six points between no load and short circuit for each of the above referenced settings. All measurements taken shall be recorded on the test report.

4.5.2.2 Volt-ampere measurements shall be taken at machine settings corresponding to minimum specified current at 20 volts and maximum specified current at rated voltage.

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4.5.3 Current adjustment measurements. -

4.5.3.1 With a variable resistance load continually adjusted to maintain rated voltage at the machine terminals, measure the current output at each machine setting. Record the measured values together with the machine settings in the test report.

4.5.3.2 Tests shall be conducted to show conformance with 3.3.16.2. The initial current and the current at the end of the six minute run shall be recorded on the test report.

4.5.3.3 With a variable resistance load adjusted to maintain rated voltage at the machine terminals, measure the output (amperes) at machine settings corresponding to minimum specified, rated and maximum specified currents.

4.5.4 Practical welding tests. - Tests to determine suitability of operation shall be conducted by a qualified arc welding operator (qualified in accordance with Standard MIL-STD-248). These tests shall include the following which shall be conducted in such manner that will not allow any unusual technique not commonly taught to or proficiently used by the average welding operator.

(a) For 200 ampere size. - Welding in all positions with 1/8 and 3/16 inch electrodes conforming to Specifications MIL-E-986 and MIL-E-15599.

(b) The steel plates used in conducting welding tests with electrodes conforming to Specifications MIL-E-986 and MIL-E-15599 shall be as follows:

(1) For Specification MIL-E-986 electrodes. - Black (uncoated) high tensile (grade HT) steel conforming to Specification MIL-S-16113.

(2) For Specification MIL-E-15599 electrodes. - High tensile grade HT steel conforming to Specification QQ-S-692 or class B of Specification QQ-S-693.

(c) The following information in connection with practical welding tests shall be recorded on the test report:

(1) Electrode size, type, class and applicable electrode specification for each test.

(2) The thickness and applicable specification number for the steel plates used to conduct each test.

(3) Position of welding.

(4) Appearance of each fillet.

(5) Any welding difficulties encountered during each test.

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4.5.5 Temperature rises. -

4.5.5.1 The temperature rise tests shall be made only on the completely assembled welding machine.

4.5.5.2 Conditions of tests. - The temperature rises shall be determined under the following conditions:

(a) Continuous operation of the welding machine on the rated load setting with a load equal to rated voltage and current on duty cycles of 6 minutes at rated output and 4 minutes at no load until all temperatures are ultimate.
(b) Continuous operation of the welding machine with 20 volts at the terminals and with maximum current obtainable at 20 volts on the lowest current setting under a duty cycle of 6 minutes at maximum current and 4 minutes at no load until all temperatures are ultimate.
(c) Ultimate temperature is defined as the condition where the component temperatures do not increase when at least three consecutive readings are taken at not less than 10 minute intervals.
(d) Blower motors shall be in continuous operation during temperature rise tests if such motors are incorporated in the design of the welding machine under test.

4.5.5.3 Measurement of ambient temperature. -

4.5.5.3.1 The welding machine may be tested at any convenient room temperature above 10°C. The temperature rise for any set of operating conditions shall be considered independent of ambient temperature if such temperature is between 10°C. and 50°C.

4.5.5.3.2 The ambient room temperature shall be measured by means of two or more thermometers placed at different points around and on a level with the welding machine and at a distance of 3 to 6 feet therefrom. The room temperature thermometers shall be inserted in oil filled cups of not less than 1 inch external diameter and 2 inches high, and shall be protected from drafts and from abnormal heat radiation. The value of the ambient temperature for the temperature rise tests shall be the mean of the readings of the thermometers (placed as stated above) then at four intervals of time during the last quarter of the temperature test.

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4.5.5.3.3 The conditions of testing room shall be such that the room temperature does not vary appreciably during the test. A variation of 10°C., or over, during the test shall not be exceeded. If room temperatures are very irregular during the test, or changes rapidly at the end of the test, the test shall be repeated.

4.5.5.4 Methods of measuring temperature. - Temperatures shall be measured by thermometer, thermocouple or the resistance methods in accordance with Specification MIL-E-917.

4.5.5.5 Temperature test data. - During each temperature test the following data shall be taken at regular intervals at the end of the 6 minute periods when the machine is delivering the loads specified in 4.5.5.2 (a) and 4.5.5.2 (b) herein, and recorded on the test report.

(a) Room ambient temperature.
(b) Ambient temperatures of the cooling air supplied to each rectifier stack, capacitor, resistor, rheostat and the blower motor.
(c) Temperatures of all transformer and reactor windings.
(d) Maximum operating temperature for the plates in each rectifier stack.
(e) The temperature of cooling air being expelled from the welding machine.

4.5.6 Noise. - Observation shall be made of the magnitude of noise being emitted from the welding machine during the progress of testing to determine that the machine is free from any degree of noise comparably greater than that inherent in the given design and size of equipment.

4.5.7 Short circuit and recovery tests. - Simultaneous current-time and voltage-time relationships for these tests shall be recorded by means of an oscillograph. Starting with open-circuit, the welding machine terminals shall be short circuited by means of a contactor with blow-out coil. After the short-circuit current has become steady, the short circuit shall be opened through a resistive load which will give normal current for that setting. After the normal current has become steady, the welding machine terminals shall be short circuited by means of the contactor and blow-out coil. The time of recovery shall be the time elapsed from the opening of the short-circuit until the current is within 5 percent of the normal current for that setting. Short circuit and time of recovery characteristics shall be determined for at least the following settings:

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(a) The setting corresponding to rated output.
(b) The setting corresponding minimum specified output current at 20 volts.
(c) At least one other setting within the specified range.

During these tests the oscillograph film shall travel at sufficient speed to distinctly show the maximum and minimum voltages and currents and the time relationships involved. All measurements shall be recorded on the test report.

4.5.8 Efficiency. - The input power shall be measured at the input terminals of the welding machine at rated voltage and frequency. The output shall be measured at the welding terminals of the machine when the machine is delivering loads corresponding to 125, 100, 75, 50 and 25 percent of rated output. The efficiency measurements shall be made immediately after the temperature tests at rated voltage and current while the machine is at approximately normal operating temperature. The percent efficiency shall be 100 times the measured output divided by the measured input. All efficiency measurements shall be recorded on the test report.

4.5.9 Weight. - The weight of the welding machine complete with all component parts shall be taken and recorded on the test report.

4.5.10 Inclined operation. - Inclined operation tests shall be conducted with the welding machine inclined 30 degrees from the normal horizontal position in each of the four directions. Under each of these four positions, the welding machine shall be run at no load for a period of not less than one-half hour. During the progress of these tests, the mechanical balance of rotating components shall be equal to the balance in the normal horizontal position, there shall be no grinding of bearings and lubrication shall be satisfactory. Further, during the progress of these tests the operation of current adjustments shall be observed to determine that the control system can be easily adjusted (no sticking or binding shall be experienced) when the machine is in the inclined position.

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4.5.11 Special devices. - Special devices shall be tested as specified (see 6.1).

4.5.12 Dielectric tests. - The dielectric test shall be conducted after the tests required by 4.5.1 through 4.5.11 have been completed. This test shall be conducted on the completely assembled welding machine. The test voltage shall be of the magnitude and shall be applied as specified in 3.3.21.2 for a period of one minute. The dielectric test voltage and the circuit or circuits to which it is applied shall be recorded on the test report.

4.5.12.1 Measurement of test voltage. - The voltmeter method shall be used. In measuring the voltage with a voltmeter, the instrument shall derive its voltage from the high voltage circuit directly or through an auxiliary ratio transformer or by means of a voltmeter coil placed in the testing transformer.

4.5.13 Insulation resistance (cold). - This test shall be made before the dielectric tests. Prior to application of the test voltage, the circuits shall be thoroughly discharged by connecting each circuit to the frame. Circuits of equal voltage above ground shall be connected together. Circuits or groups of circuits of different voltage above ground shall be separated. Separate measurements shall be made on circuits which are of different voltage above ground. Insulation resistance shall be measured with an insulation resistance indicating ohmmeter conforming to the requirements of type B or G of Specification MIL-O-16485. The time of test voltage application shall not be less than 60 seconds. All readings, the applicable circuit for each reading and the temperature at which these readings were taken shall be recorded in the test report.

4.5.14 No-load input power. - The input power to the welding machine shall be measured when the machine is operating at no load and at rated output.

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4.5.15 Parallel operation. - Two welding machines shall be operated in parallel using a resistive load and at a machine setting corresponding to rated output of the machine and at least one other setting within the specified range. At least 11 readings (6 readings with increasing load and 5 readings with decreasing load) shall be taken for each machine setting by varying the total load on machines in parallel. The load shall be varied at approximately equal intervals between loads corresponding to 10 percent and 125 percent of two times the normal current for each machine setting involved. All setttings on each machine shall be accomplished prior to conducting each paralleling test. Following the initial setting for each paralleling test, there shall be no readjustment of the machine setting for the remainder of each test. Measurements of total load (amperes), the load (amperes) supplied by each machine in parallel and the setting on each machine shall be recorded on the test report.

4.5.16 HI impact shock tests. -

4.5.16.1 The class HI shock tests shall be conducted in accordance with the type A requirements of Specification MIL-S-901. Tests on the machine shall be conducted on the lightweight or the medium-weight shock machines. The machine shall be tested, on alternate blows, in each of two operating conditions namely; running and standing. The machine shall be tested in the running condition when it is delivering rated voltage and current. During tests in the standing condition, the machine input terminals shall be connected to 440 volt, 3 phase, 60 cycle power source with the switch or circuit breaker on the machine in the "off" position.

4.5.16.2 Inspection after shock test. -

4.5.16.2.1 Upon completion of the shock test, the welding machine shall be carefully inspected and tested to determine the extent of damage to the electrical and mechanical parts. Any noticeable increase or decrease in noise, vibration and the ease of adjusting controls should be investigated and its cause determined.

4.5.16.2.2 The operating characteristics of the machine as obtained by the tests in 4.5.2 and 4.5.3 shall be spot checked to determine that thay have not changed by more than 10 percent.

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4.5.16.2.3 The machine shall be operated under load until the operating temperatures of parts correspond to the values attained at the completion of the temperature tests specified in 4.5.5.2 (a). After the above operating temperatures have been attained, the welding machine shall be disconnected from its power source and load, and a dielectric strength test shall be conducted in accordance with the requirements of 4.5.12 except that the applied voltage shall be 1000 volts.

4.5.16.2.4 Following the check tests in 4.5.16.2.1 through 4.5.16.2.3, the welding machine shall be disassembled and inspected for damage. The extent of disassembly shall be conducted to a point where the condition of the machine and its parts can be easily observed.

4.5.16.3 The effect of shock as observed during the tests and inspection required by 4.5.16.1 through 4.5.16.2.4 shall be carefully observed and recorded on the test report.

4.5.16.4 Causes for rejection. - The following shall be cause for rejection of the welding machine under the high impact shock test:

(a) Breakage of any part.
(b) Appreciable distortion or dislocation of any part.
(c) Nonconformity with the tests specified in 4.5.16.2.2 and 4.5.16.2.3.
(d) Starting of the machine from the deenergized condition or stopping of the machine from the energized condition from any shock blow.
(e) Damage to the extent that proper operation or control of the machine cannot be exercised.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Basic equipment. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

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5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.1.3.1 Technical manuals shall be placed in one envelope marked "Instruction Data". The envelope shall be attached securely to each machine.

5.1.2 Onboard repair parts. - Repair parts shall be preserved and packaged level A, packed level C and marked levels A and C respectively in accordance with the requirements of Specification MIL-P-16298 or MIL-E-17555, as applicable.

5.2 Domestic shipment, storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Basic equipment. - The welding machine shall be preserved and packaged level A or C as specified in accordance with Specification MIL-P-16298.

5.2.1.1.2 Repair parts. - Repair parts shall be preserved and packaged in accordance with level A requirements of Specification MIL-P-16298 or MIL-E-17555, as applicable.

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5.2.1.2 Packing. -

5.2.1.2.1 Basic equipment. -

5.2.1.2.1.1 Levels A and B. - The basic equipment shall be packed level A or B as specified in accordance with Specification MIL-P-16298. Disassembled parts and accessories shall be packed in fiberboard boxes conforming to Specification PPP-B-636 and secured within the equipment container.

5.2.1.2.2 Repair parts. - Repair parts shall be packed level A or B, as specified in accordance with Specifications MIL-P-16298 or MIL-E-17555, as applicable.

5.2.1.3 Technical manuals. - Technical manuals shall be packaged and packed for the level specified in accordance with Specification MIL-M-15071.

5.2.1.4 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked for shipment in accordance with the applicable packaging and packing specifications.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Special devices (see 3.4).
(d) Accessories required (see 3.5).
(e) Repair parts (onboard) required (see 3.8).
(f) Number of technical manuals required if other than as specified (see 3.11).
(g) Test requirements for special devices (see 4.5.11).
(h) Preservation, packaging, packing and marking if other than as specified in 5.1 (see 5.2).

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6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL 19285, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2.1).

6.2.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specification, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacturer, use, or sell any patented invention that may in any way be related thereto.

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